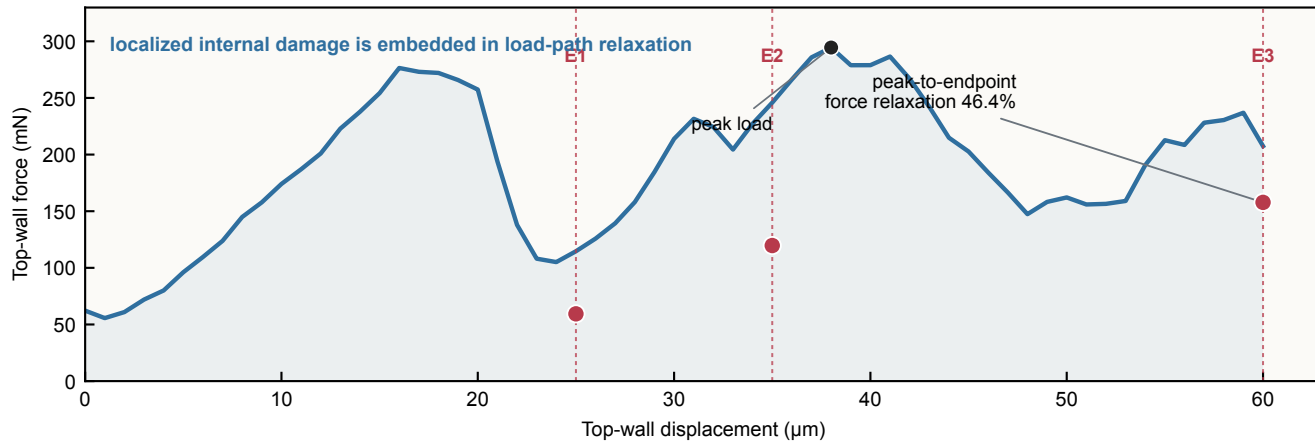
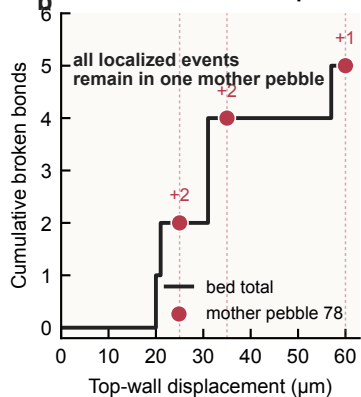


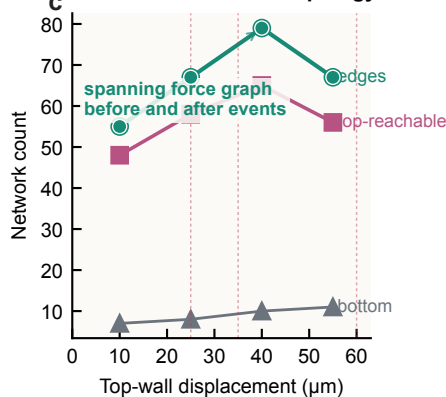
a macro-response chronology



b localized bond-loss sequence



c native force-network topology



d damage remains in an upper-bed mother pebble

